

The Role of Regenerative Suspension in Energy-Efficient and Comfortable Electric Vehicle Driving

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ABSTRACT

This study investigates the energy recovery potential of regenerative suspension systems in battery-electric vehicles and their direct impact on ride comfort under diverse urban driving conditions. Moving beyond simplified estimation methods, a comprehensive electromechanical modeling framework was developed in MATLAB/Simulink. This framework integrates a validated 2-DOF quarter-car model with an explicit generator dynamics and a longitudinal BEV model, ensuring that the regenerative damping force is directly coupled with the vehicle's motion. The longitudinal model was validated against NEDC data with less than 1% deviation, and the suspension's electromechanical feedback was found to be consistent with established literature. Simulations were conducted over a speed range of 2–50 km/h on semi-circular and trapezoidal speed bumps, as well as on stochastic ISO 8608 Class A and Class B road profiles. The results reveal that energy recovery is highly sensitive to both vehicle speed and road roughness. For discrete road inputs, maximum energy values of 0.406 Wh and 0.659 Wh were recovered for semicircular and trapezoidal bumps, respectively. Under continuous excitation on ISO Class B roads, the system demonstrated a significant recovery potential of up to 70.4 Wh per kilometer at low speeds, representing a meaningful supplementary energy source. A critical inverse relationship was observed between vehicle speed and harvested energy; as speed increased to 50 km/h, the reduced interaction time between the suspension and road irregularities led to a sharp decline in recovery efficiency. Furthermore, the ride comfort analysis, evaluated per ISO 2631 criteria, highlighted a distinct trade-off: while lower speeds favored both superior energy recovery and passenger comfort, higher speeds required a balance between energy gains and increased vertical accelerations. Overall, this research demonstrates that integrating physics-based electromechanical coupling provides a more accurate and promising outlook for regenerative suspension systems in realistic driving scenarios.

Keywords: Battery electric vehicle; Energy recovery; Regenerative suspension; Vibration; System dynamics

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1. Introduction

The increase in the world population also increases energy requirements, and the more sustainable use of energy resources becomes increasingly important [1]. Although the complete elimination of fossil fuel dependence is not currently feasible, the utilization of electricity in vehicles (especially from renewable sources) is crucial for promoting resource sustainability and decreasing fossil fuel reliance [2]. Electric vehicles have significant potential to advance sustainable mobility and create a more carbon-neutral environment for future generations. It helps reduce air pollution caused by traffic, thanks to its zero exhaust emissions

enabled by its technical infrastructure. Road transport electrification, defined as the shift from conventional internal combustion engine vehicles to electric vehicles, constitutes a critical component of the 2050 net-zero scenario consistent with the Paris Agreement's 1.5 °C goal [3]. When evaluated economically, the higher energy efficiency of electric vehicles relative to internal combustion engines results in lower per-unit-distance travel costs. Despite earlier high prices, electric vehicles have become accessible to the general public due to an 85% decline in Li-ion battery costs over the past decade [4, 5]. Electric vehicles can also reduce urban noise pollution because their electric drivetrains operate more quietly than internal combustion engines, especially at low speeds in city traffic [6]. Despite these

advantages, the primary obstacle hindering the widespread adoption of electric vehicles is their limited driving range, which is directly dependent on battery capacity [7].

To provide the expected driving range, it is necessary to reduce energy consumption per distance or to install a larger-capacity battery in the vehicle. Using a large-capacity battery to increase range will increase the vehicle's load and limit passenger seating. In addition, the use of air conditioning units in electric vehicles significantly reduces driving range [8]. Although efforts are made to address the problem through technological advances in thermal issues, the decrease in driving range continues. For these reasons, driving range anxiety occurs in electric vehicle drivers [9]. Many studies are underway on energy management in electric vehicles to improve driving comfort and reduce range anxiety. In general, various methods are used to achieve energy-optimal behavior in single- or multi-energy-source vehicle systems [10]. Range extension strategies include optimal vehicle design with metaheuristic optimization methods, artificial intelligence-supported driving prediction [11], and driving behavior modeling with data analysis methods. Moreover, fuel cells that generate electricity from hydrogen are also used to supplement the battery's limited energy capacity and increase driving range.

In addition to determining the right design strategy and software support in electric vehicles, some energy recovery methods are also being investigated to increase driving range. Since the electric motor that drives electric vehicles is reversible, it can generate electricity in an external motion drive. This feature enables regenerative braking in electric vehicles. When the accelerator pedal is released while driving, the electric motor operates in regenerative braking mode, slowing the vehicle while generating electricity and charging the battery. Solar panels are mounted on electric vehicles with large exterior surfaces to generate electrical energy and store it in the battery [12]. Thanks to technological advances, energy can be generated variable sources. Energy production by connecting an electric generator to the suspension system of road vehicles is also a suitable example of energy recovery types.

Regenerative suspension is an energy recovery method that generates electricity by exploiting the vertical motion of a vehicle's suspension. Due to road irregularities such as potholes and bumps, vehicles experience vertical motion that can adversely affect ride comfort [13]. To mitigate this, suspension systems are employed to isolate the vehicle body from road-induced vertical displacements through appropriate stiffness and damping characteristics. During driving, a portion of the vertical road input is attenuated by the suspension, while the remaining motion can be converted into rotational motion via mechanical mechanisms and transmitted to an electrical generator. Similar to regenerative braking, the generator produces electrical energy from this motion. Energy recovery can be achieved by integrating generators into the suspension systems of all four wheels. For effective implementation, the generator must be compact enough to fit within the suspension assembly, capable of generating power under small suspension displacements, cost-effective, lightweight, and require minimal maintenance.

In recent years, the number of studies [14–17] aimed at accelerating the integration of regenerative suspension systems into electric vehicles and maximizing energy recovery without compromising ride comfort and road holding has been steadily

increasing. However, this quest is no longer limited to simply replacing traditional damping systems; it pushes the mechanical and electronic limits to transform each shock into a micro-scale power plant. For example, next-generation shock absorbers that leverage the highly efficient energy-harvesting capabilities of helical gear mechanisms not only improve ride comfort but also directly enhance the critical range capacity of electric vehicles [14]. However, the focus of technology is not limited to the wheel-to-road contact. Modern approaches offer holistic vibration management by even designing the inertial systems used in seat suspensions in commercial vehicles as energy recovery points [15]. Hybrid electromagnetic active suspension systems both maintain system efficiency through energy-saving control algorithms and enable self-sustainability by intelligently managing recovered energy [16]. In essence, this technological transformation [17] is turning electric vehicles from a static means of transport into a dynamic, sustainable ecosystem that converts every imperfection of the road into efficiency.

Various simulation techniques are used in system-based studies, such as regenerative suspension. While finite element models are preferred for 3D examination of systems [18, 19], strength analysis, and fatigue life determination, 1D simulation models are used in system dynamics studies [20]. Simulation models inspired by real models yield time and cost savings across different studies. They can provide an examination of system dynamics and various optimizations. For example, different road and speed scenarios can be easily tested with the vehicle longitudinal dynamic model, cornering ability can be examined with the vehicle lateral dynamic model, and comfort examination can be performed under different road conditions with the vehicle vertical dynamic model. It should be known how accurately the virtual model that will provide output to the study represents the real conditions. Model accuracy conditions can be determined by comparing against a well-known virtual model or by repeating a basic real-world scenario. The metrics examined in the comparison phase should be of sufficient quality to accurately reflect the system's behavior. While metrics such as position, speed, acceleration, and energy consumption are examined in longitudinal vehicle dynamics, metrics such as position, force, and acceleration are examined in suspension systems. Although in many studies the virtual model's accuracy is expected to be 100% or close to it, a model whose accuracy has been determined at any rate is suitable for the study. At this point, the important thing is to know the virtual model's accuracy rate. The model's accuracy will be taken into account in evaluating the obtained outputs.

The existing literature extensively covers the system dynamics and potential energy-harvesting capabilities of regenerative suspension systems. However, this study diverges from conventional research by integrating a comprehensive virtual electric vehicle model to quantify the actual contribution of harvested energy to total vehicle consumption under specific operational scenarios. Beyond simple energy recovery, the analysis identifies the optimal speed for energy-efficient driving while simultaneously evaluating the critical trade-off between power conservation and passenger comfort. By establishing this balance, the study provides a more holistic perspective on how regenerative systems impact real-world electric vehicle performance.

2. Methodology

The energy recovery potential of regenerative suspension was investigated using virtual models. For this purpose, a virtual electric vehicle model and a 2-degree-of-freedom (DOF) quarter-car model were created in MATLAB/Simulink R2022b. The models were correlated based on cumulative energy consumption and dynamic response. The virtual model was run over semicircular and trapezoidal bumps at different constant speeds, and the energy consumption and recovery were investigated. Since the system's dynamic response may affect passenger comfort, the comfort level was defined based on the vertical acceleration results.

The layout of the virtual model is shown in Figure 1. The vehicle model provides outputs of speed, energy consumption, and distance, with inputs of speed target and road slope. The road slope is defined by the horizontal and vertical distances, which are sent as input to the vehicle model. During this process, the 2-DOF quarter-car model calculates the dynamic responses of the vehicle body and the tires. Body and tire position, suspension force and speed, body acceleration, and regenerated energy.

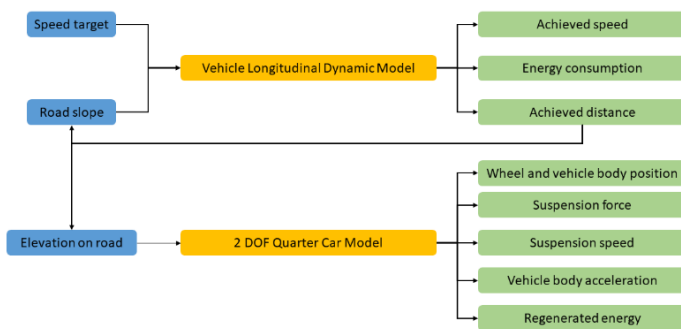


Figure 1. Virtual model layout

In addition to energy recovery, the suspension system's dynamic response also affects driving comfort. The impact that the suspension cannot absorb is transmitted to the vehicle body, potentially causing vertical movement. For this reason, an acceleration output will occur. In the studies conducted, vehicle acceleration values were used to assess driving comfort limitations and are presented in Table 1. In this study, acceleration values observed in the vehicle body according to the ISO 2631 standard and reference studies [21, 22] were evaluated using the classification in Table 1 and discussed in terms of comfortable driving.

Table 1. Acceleration comfort assessment

Acceleration (m/s^2)	Comfort level
Below the 0.3	6
Between 0.3 and 0.6	5
Between 0.5 and 1	4
Between 0.58 and 1.6	3
Between 1.25 and 2.5	2
Above the 2	1

2.1. Virtual electric vehicle model

To investigate the effect of regenerative suspension efficiency on energy consumption, two types of bumps commonly found on urban highways were used. Therefore, the virtual electric vehicle model was created based on the characteristics of a real battery electric vehicle, whose technical specifications are frequently reported in the

literature and are suitable for urban use. The technical specifications of the electric vehicle are given in Table 2 [23, 24].

Table 2. Technical specifications of electric vehicle

Vehicle Specification	Value
Vehicle type	B-hatchback
Weight	1443 kg
Vehicle front area	2.38 m^2
Drag coefficient	0.3
e-Motor maximum speed	11470 rpm
e-Motor maximum power	125 kW
e-Motor maximum torque	250 Nm
Efficiency	The efficiency map is available in the reference
Transmission ratio	9.7
Tire radius	0.33 m
Battery nominal capacity	22 kWh

The virtual vehicle model was developed in MATLAB/Simulink R2022b based on fundamental longitudinal vehicle dynamics, as detailed in the literature [20, 25]. A general view of the model is presented in Figure 3. Vehicle longitudinal mobility is governed by its ability to overcome four primary resistance forces: aerodynamic drag related to vehicle speed and frontal area, grade resistance due to vehicle mass and road slope, rolling resistance associated with tire and vehicle properties, and inertial resistance arising from acceleration. Drive torque is transmitted to the wheels through the powertrain. When the wheel torque equals the sum of these resistive forces, the vehicle maintains constant speed; if it exceeds them, the vehicle accelerates; if it is lower, the vehicle decelerates. In electric vehicles, torque generation depends on the battery current demand, whereas in internal combustion engines, it depends on the fuel supplied to the cylinders. The overall structure of the battery-electric virtual vehicle model used in this study is shown in Figure 2.

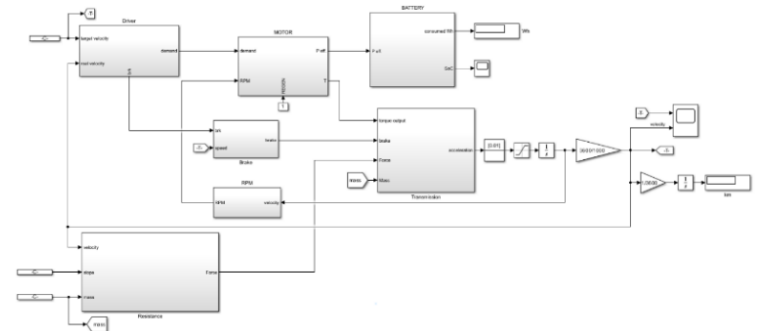


Figure 2. Electric vehicle model in Simulink

The energy consumption of the real battery electric vehicle model according to the New European Driving Cycle (NEDC) is reported as 0.1175 kWh/km [26]. To determine the accuracy of the energy consumption output of the virtual vehicle model created according to the vehicle's technical specifications, the same driving cycle was run under the same conditions as the real one, and the energy consumption value was recorded. The energy consumption of the virtual electric vehicle model over the 11 km NEDC route was 0.1167 kWh/km. According to this result, the virtual vehicle model error is less than 1% for cumulative energy consumption. The vehicle speed and energy consumption graphs resulting from the NEDC driving cycle run of the virtual vehicle model are given in Figure 3.

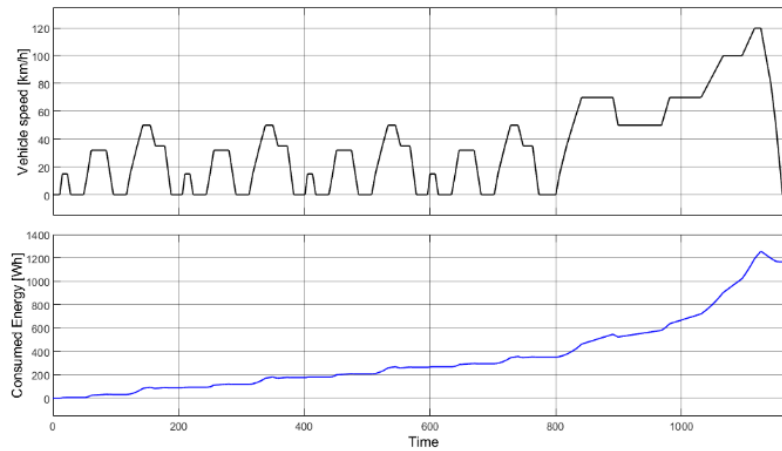


Figure 3. Vehicle speed and energy consumption correlation of the virtual vehicle model on NEDC

2.2.2. DOF passive suspension quarter car model

In the energy recovery examination of regenerative suspension, a quarter car model, which is frequently preferred in literature [27–30] due to its simplicity and compatibility with real systems, has been designed. The real electric vehicle's traction system is on the front wheels, and the quarter-vehicle model has been designed with the front wheels to be compatible with the real model. The quarter-vehicle model includes the mass of the vehicle body, wheels, springs, dampers, tire springs, and tire dampers. Similar to the displacement that occurs during the driving of a real vehicle, the vertical elevation change on the road surface in the virtual model is first damped by the wheel's spring and damper characteristics. The undamped displacement is transferred to the spring and damper part of the vehicle. In the third stage, the final displacement value is transferred to the vehicle body. The virtual 2-DOF quarter-vehicle model is shown in Figure 4.

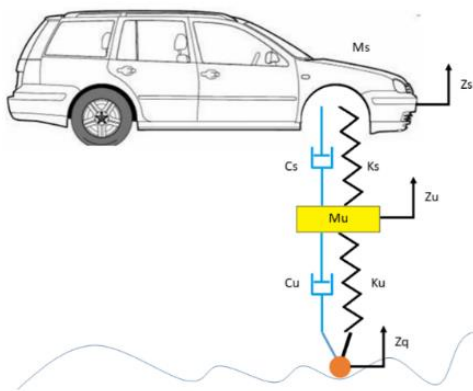


Figure 4. Quarter car model

The quarter-car model with shared dynamic equations was developed in MATLAB/Simulink R2022b. Like in real-world conditions, this system is initialized with the vertical displacement input to the unsprung mass. The damped first impact is transferred to the spring and damper of the sprung mass side. The general visual of the virtual model is shown in Figure 5. The required parameter values in the virtual model are taken from the literature [31, 32].

When the mass distribution in a real battery-electric vehicle is assumed equal, the mass per wheel is quite close to the value in the reference study. The values of the spring and damper parameters were preferred in this study because they were considered suitable for the mass per wheel and vehicle used in the city. The parameter values of the model are: sprung mass (m_s) 365.4 kg, unsprung mass (m_u) 43 kg, suspension stiffness (k_s) 24000 N/m, suspension damper (c_s) 2126.4 Ns/m, tire stiffness (k_u) 350000 N/m, tire damper (c_u) 1000 Ns/m.

For the numerical solution of the differential equations, the initial conditions were defined as $z_s(0)=0$, $\dot{z}_s=0$, $z_u(0)=0$, and $\dot{z}_u=0$, assuming the vehicle starts from a state of static equilibrium.

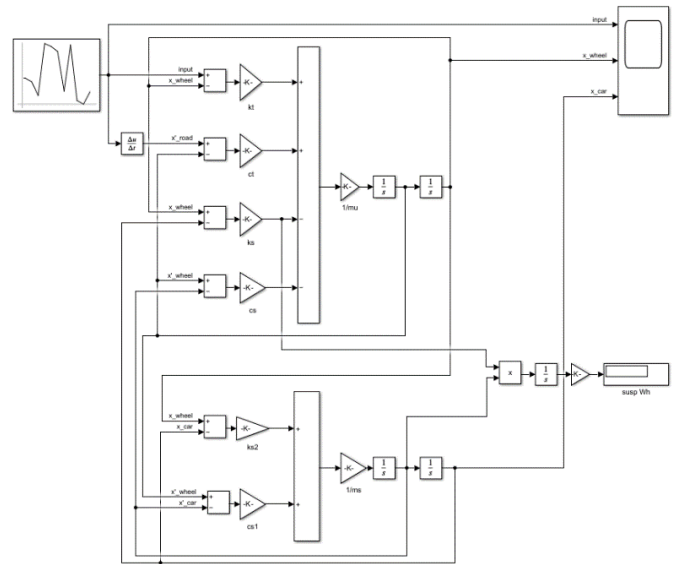


Figure 5. Quarter car model in MATLAB/Simulink

To verify the accuracy of the quarter-car model's dynamic outputs, the outputs generated from the inputs in reference [31] were compared. It was observed that the displacement values obtained for the ramp and step inputs on the wheel and the vehicle body were quite consistent with the reference. The visual results for the quarter-car model dynamic correlation are shown in Figure 6.

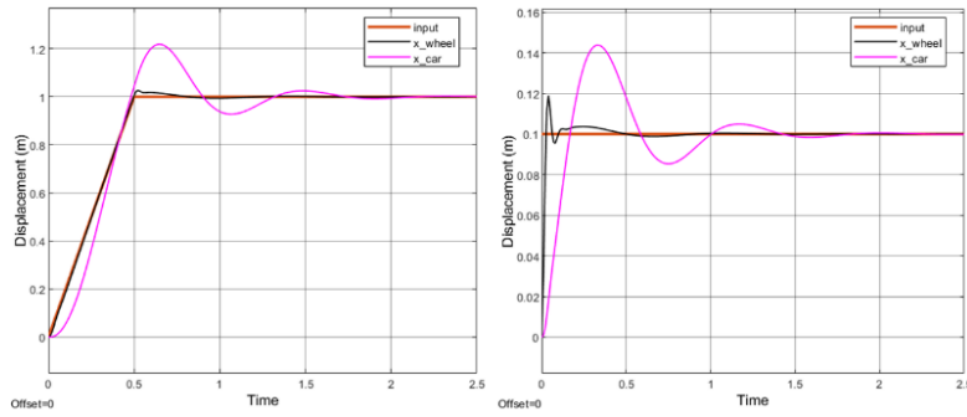


Figure 6. Dynamic response correlation of quarter car model (a) ramp input, (b) step input

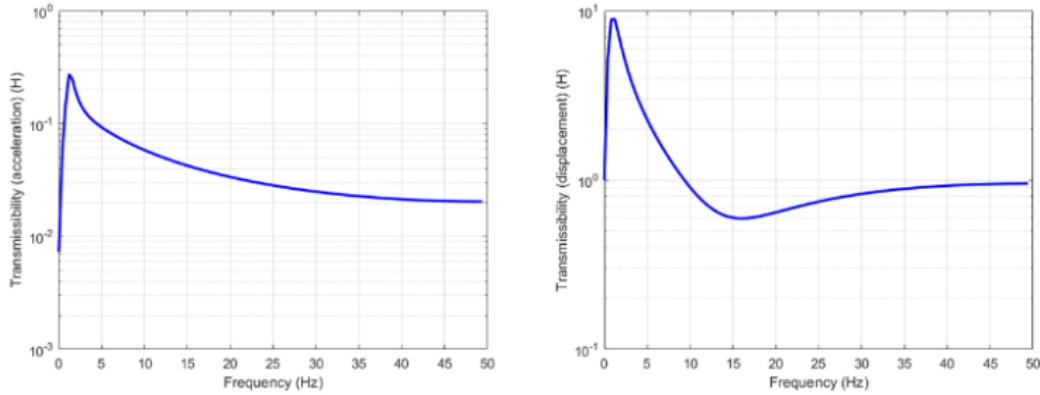


Figure 7. Transmissibility curves of the system (a) acceleration ratio, (b) displacement ratio

The energy recovery process in the suspension system is conceived as an electromechanical conversion cycle where mechanical vibrations are directly related to electrical parameters. The process begins with the transmission of the suspension relative speed ($\dot{z}_s - \dot{z}_u$), triggered by irregularities in the road surface, to the generator rotor via a high transmission ratio (G); this mechanical input induces a back-EMF (V_{emf}) voltage in the generator windings, proportional to the speed, according to Faraday's law of induction. This voltage induces a current (I) through the circuit impedance, which is the sum of the generator's internal resistance (R_{int}) and the external load resistance (R_L). At this point, the harvested net power (P_{net}) is defined by the electrical dissipation across the load resistance; while the fundamental element ensuring the physical consistency of the system is the feedback of the resistive torque created by the flowing current on the generator, according to Lenz's Law, back to the suspension as a 'regenerative damping force' (F_{reg}). Thus, the bidirectional interaction (coupling) between mechanical damping effect and electrical energy generation is modeled within an integrated structure in accordance with the physical framework proposed in reference [33]. The parameter values of the method are: $G=450$, $k_e=72$, $k_t=25$, $R_{int}=5.3$, $R_L=15$, efficiency (η_m)=0.9. The methodology of energy harvest is in Eq. (1-5).

$$V_{emf}(t) = k_e \cdot G \cdot (\dot{z}_s(t) - \dot{z}_u(t)) \quad (1)$$

$$I(t) = \frac{V_{emf}(t)}{R_{int} + R_L} \quad (2)$$

$$F_{reg}(t) = \frac{k_t \cdot I(t) \cdot G}{\eta_m} \quad (3)$$

$$P_{net}(t) = I(t)^2 \cdot R_L \quad (4)$$

$$E_{regen} = \int_{t_0}^{t_f} P_{net}(t) dt \quad (5)$$

Transmissibility shows how a system responds to an impact as a function of frequency. It is often used in vibration analysis and is an important parameter for evaluating the dynamic response of quarter car models, suspension systems, and mechanical systems. In the study, the dynamic behavior of the quarter-vehicle model, which was correlated with the reference, was obtained using a step input, and the transmissibility graphs were generated. Calculations were made on the ratios of the displacement and acceleration values occurring in the wheel and vehicle body. In obtaining the transmissibility curve, the sampling frequency was calculated over the sampling time. Cooley-Tukey type Fast Fourier Transform (FFT) was applied [34]. The frequency axis was created by considering the Nyquist Theorem [35]. Finally, the transmissibility was obtained using the data type (acceleration or displacement) for the wheel and the vehicle body. Transmissibility values were written on the graph according to the frequency values in Figure 7.

As shown in the transmissibility graphs in Figure 8, the transmissibility reached its maximum at 1.19 Hz. In the transmissibility graph plotted as a function of displacement ratio, the

highest damping was observed at approximately 15 Hz.

The virtual model, which includes the virtual electric vehicle and quarter vehicle model designed in the study, was passed over two different hump models [36], semicircular and trapezoidal, at constant speeds of 2 km/h, 5 km/h, 10 km/h, 20 km/h, 30 km/h, 40

km/h, and 50 km/h. During this time, energy consumption in the virtual vehicle model and the energy recovered in the regenerative suspension model were measured. The hump models used in the study are shown in Figure 8.

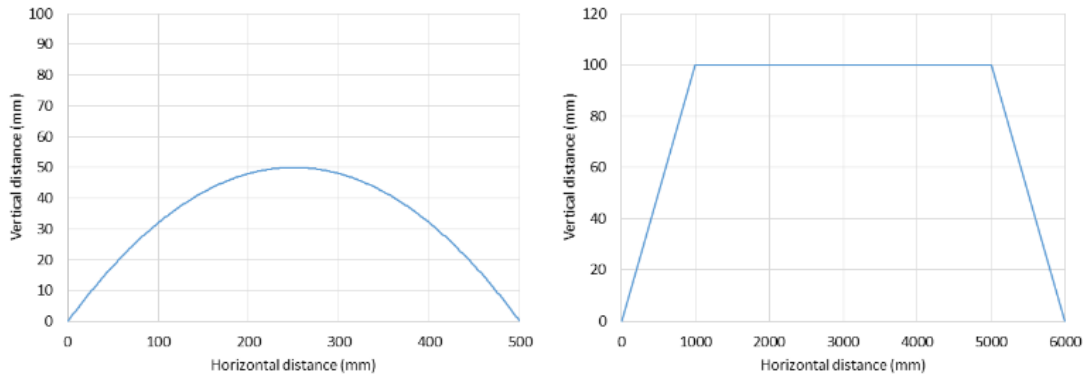


Figure 8. Cross sections of hump models (a) semi-circle, (b) trapezoidal

2.3. ISO 8608-based road roughness representation

In this study, road surface irregularities are represented using the ISO 8608 standard [37], which characterizes road roughness in terms of the spatial power spectral density (PSD) of vertical road displacement. This approach enables a standardized, reproducible, and physically meaningful description of road-induced vertical excitation, independent of specific vehicle configurations.

According to ISO 8608, the spatial PSD of the road profile is expressed in Eq. (6).

$$G_q(n) = G_q(n_0) \left(\frac{n}{n_0} \right)^{-w} \quad (6)$$

Where n denotes the spatial frequency (m⁻¹), n_0 is the roughness coefficient at the reference spatial frequency n_0 is 0.1 m⁻¹, and w is the waviness exponent, typically taken as 2 for paved roads. The roughness coefficient defines the ISO road class and directly governs the amplitude of the resulting surface profile. To generate a realizable road profile in the spatial domain, the PSD description is transformed into a deterministic multisine representation with random phase components. The vertical road displacement as a function of horizontal distance x is given by Eq. (7).

$$z(x) = \sum_{k=1}^N A_k \cos(2\pi n_k x + \theta_k) \quad (7)$$

Where n_k is the k th discrete spatial frequency, Δn is the frequency increment, N is the number of harmonics, and θ_k represents uniformly distributed phase angles in the interval $[0, 2\pi]$. The frequency resolution is defined as $\Delta n = (n_{\max} - n_{\min}) / (N - 1)$.

The amplitude of each harmonic component is calculated directly from the target PSD as in Eq. (8).

$$A_k = \sqrt{2G_q(n_k) \Delta n} \quad (8)$$

Each harmonic amplitude is selected to ensure statistical convergence of the synthesized road profile to the target ISO 8608 spectrum. The chosen spatial frequency range captures both long-

wavelength road undulations and short-wavelength irregularities relevant to suspension excitation. Very low frequencies are excluded to avoid unrealistically large quasi-static displacements, while the upper frequency limit is set high enough to retain all dynamically significant content.

To ensure repeatability, random phase angles are generated using a deterministic pseudo-random number generator with a fixed seed. This guarantees identical road profiles for a given road class across all simulations, enabling consistent comparisons between suspension configurations and operating conditions.

The road profile is defined as a function of longitudinal distance, with inputs in kilometers and internally converted to meters. The resulting vertical displacement, expressed in meters, is applied directly as a base excitation at the tire–road interface of the quarter-car model. Longitudinal slope effects and forward vehicle dynamics are neglected, allowing the analysis to focus solely on vertical vibrations induced by road roughness.

By combining ISO 8608 spectral definitions with a deterministic multisine approach, the methodology provides a standardized and reproducible framework for road excitation, enabling fair comparisons across road classes while remaining compatible with time-domain vehicle dynamics simulations.

In passenger vehicle suspensions, the relative motion between sprung and unsprung masses is limited by suspension travel and bump-stop constraints. Typical systems allow a total vertical travel of 120–160 mm, comprising approximately 60–90 mm of compression and 40–70 mm of rebound. Accordingly, the suspension working space is limited to ± 0.08 m in this study, reflecting realistic packaging constraints for road vehicles [38].

Figure 9 illustrates the generated road displacement profiles corresponding to ISO 8608 road roughness Classes A–F, obtained using the spatial PSD–based multisine approach. The vertical displacement amplitude increases progressively with the ISO class, reflecting the higher roughness levels defined in the standard. It is observed that the Class A and Class B profiles remain within the admissible vertical excitation bounds of typical passenger car suspension systems, whereas higher classes (C–F) introduce displacement amplitudes that exceed realistic operating limits for conventional passenger vehicles. Therefore, in subsequent

simulations, only ISO 8608 Class A and Class B road profiles are used as vertical wheel excitations to ensure physical consistency with suspension travel constraints and to represent real-world driving conditions.

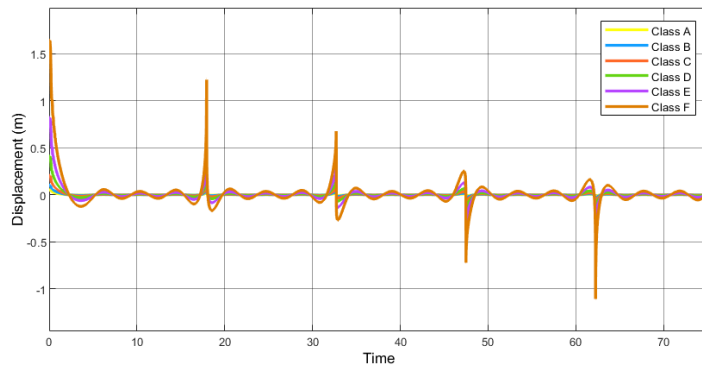


Figure 9. Vertical road displacement profiles generated according to ISO 8608

3. Results and Discussions

In this study, which examines the effectiveness of regenerative suspension, a virtual vehicle model inspired by a compact battery-electric vehicle suitable for urban use is created and combined with a quarter-vehicle model. The virtual system, verified with reference studies, is passed over semicircular and trapezoidal humps at constant speeds of 2 km/h, 5 km/h, 10 km/h, 20 km/h, 30 km/h, 40 km/h, 50 km/h, and the dynamic responses of the regenerative suspension, energy consumption, and recoveries are obtained. Table 3 shows the outputs obtained as the virtual model passes over the semicircle and at the end of the 1 km drive. As expected, the high speed resulted in higher energy consumption. The energy gained from the suspension decreased at speeds above 5 km/h. The decrease in energy gain with increasing speed in the semicircular hump condition is consistent with the literature [39]. The abbreviations in Table 3, as V, T, Ec, Eg, Y, and C, are vehicle speed (km/h), duration (s), consumed energy (Wh), gained energy (Wh), maximum vertical displacement (m), and comfort level, respectively.

Table 3. Results of the crossing of the semicircular hump

V	2	5	10	20	30	40	50
T	1813	722.5	361.2	181.4	122	92.56	75.18
Ec	29.63	30.65	34.47	49.35	73.8	107.6	150.4
Eg	0.406	0.341	0.338	0.143	0.157	0.167	0.175
Y	0.060	0.056	0.036	0.020	0.014	0.011	0.008
C	3	1	1	1	1	1	1

Figure 10 shows the displacement of the vehicle body (m_s) when passing over a semicircular hump at different speeds. As the speed of passage over the semicircular hump increased, the vehicle body's vertical displacement decreased. However, as the vertical acceleration increased, ride comfort decreased. Although the height of the top of the semicircular hump above the ground was 0.05 m, the vertical displacement of the vehicle body was higher than this value at low speeds. Taking into account the height of the highest point of the semicircular hump from the ground, the vehicle body is displaced by a maximum of 20% relative to the hump.

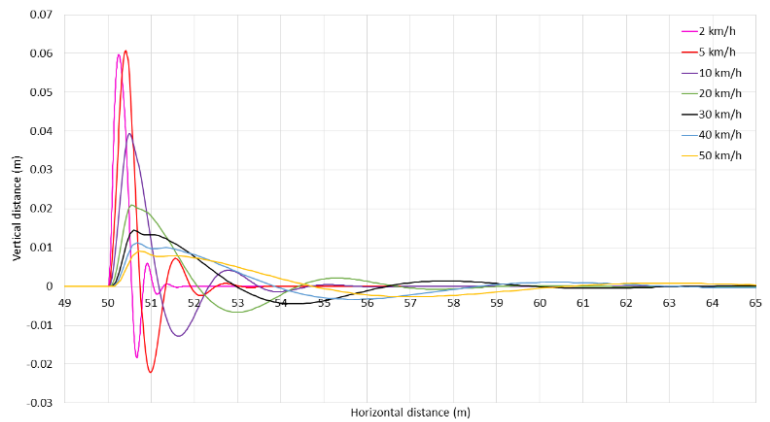


Figure 10. Vehicle body position on semicircle hump

Table 4 shows the outputs of the virtual vehicle at the specified constant speed values for the trapezoidal bump crossing and 1 km drive. The highest energy gain rate was obtained at 20 km/h in the crossing of this bump type. However, as in the semi-circle, the most comfortable driving occurred at low speeds.

Figure 11 shows the vertical displacement values of the vehicle body during the passage over the trapezoidal hump. The displacement of the vehicle body on the vertical axis in the trapezoidal hump condition compared to the semicircular hump was approximately similar at speeds of 20 km/h and above. Given that the highest point of the trapezoidal hump is 0.1 m above the ground, the vehicle body has been displaced by at most 40% relative to the hump.

Table 4. Results of crossing the trapezoidal hump

V	2	5	10	20	30	40	50
T	1813	722	361	181	122	92.5	75.1
Ec	29.6	30.6	34.4	49.3	73.8	107	150
E	0.63	0.659	0.564	0.411	0.344	0.314	0.304
g	2						
Y	0.10	0.108	0.126	0.135	0.137	0.138	0.139
		4	7	9	8	6	1
C	5	4	2	1	1	1	1

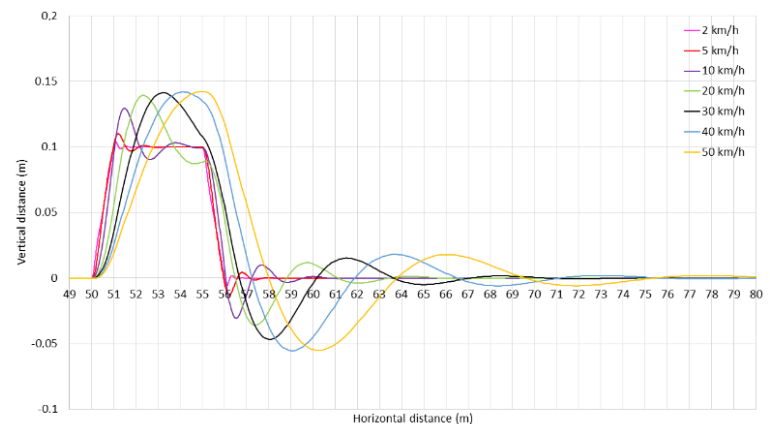


Figure 11. Vehicle body position on trapezoidal hump

Figure 12 illustrates the time-domain responses of the quarter-car suspension model subjected to ISO 8608 Class A and Class B road excitations. While both profiles remain within the physical suspension travel limits, the Class B excitation induces noticeably

higher suspension deflections and velocity levels compared to Class A. This behavior is consistent with the increased power spectral density specified by ISO 8608 for rougher road classes. Despite the higher excitation severity, the system maintains stable dynamic behavior, indicating that the suspension operates within its functional range for both road classes.

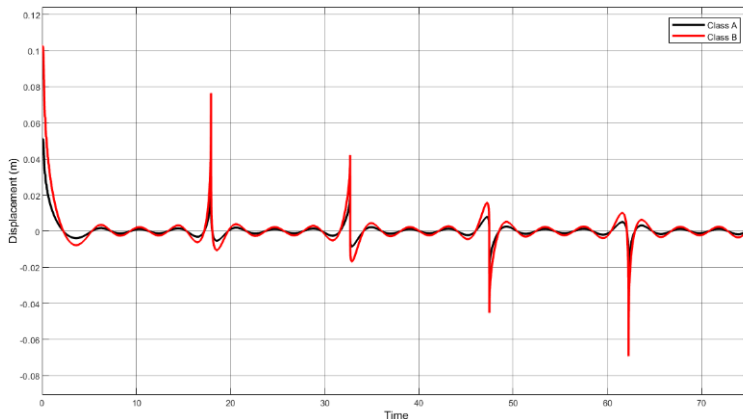


Figure 12. Wheel excitation according to ISO 8608

The results for the ISO 8608 Class A road profile, as detailed in Table 5, provide a critical baseline for energy trade-off on high-quality, smooth surfaces. Over a constant distance of 1 km, a direct, non-linear relationship is observed between vehicle speed and energy demand; specifically, as velocity increases from 2 km/h to 50 km/h, energy consumption increases from 29.63 Wh to 150 Wh. The energy regeneration was decreasing from 57.1 to 3.95 Wh. This suggests that on smoother "Class A" roads, the energy harvested from surface irregularities accounts for significant fraction of the total energy required for propulsion. Furthermore, the maximum vertical displacement decreases from 0.054 m to 0.027 m as speed increases, indicating that the suspension system experiences reduced excitation at higher speeds on this specific profile.

Table 5. Results of ISO 8608 Class A

V	2	5	10	20	30	40	50
T	1813	723	361	181	122	93	75
E	29.6	30.6	34.5	49.4	73.8	107	150
c							
E	57.1	29.7	10.5	5.52	6.16	5.45	3.95
g							
Y	0.054	0.044	0.039	0.034	0.0314	0.0294	0.0275
	5	0	2	5	5	4	4
C	1	1	1	1	1	1	1

The data presented in Table 6 illustrate a clear correlation between vehicle velocity and energy dynamics on a Class B road profile. As speed increases from 2 km/h to 50 km/h, the energy consumed per kilometer shows a significant upward trend, rising from 29.63 Wh to 150 Wh. Conversely, the energy regeneration follows an inverse trajectory, decreasing from 70.4 to 5.30 Wh. The maximum vertical displacement decreases from 0.097 m to 0.051 m as velocity increases, suggesting that higher speeds may induce a "smoothing" effect or more efficient damping of road irregularities.

Table 6. Results of ISO 8608 Class B

V	2	5	10	20	30	40	50
T	1813	723	361	181	122	93	75
E	29.6	30.6	34.4	49.3	73.8	107	150
c							
E	70.4	38.4	12.9	7.48	8.83	6.68	5.30
g							
Y	0.097	0.084	0.075	0.065	0.059	0.055	0.051
	58	82	28	44	30	80	70
C	1	1	1	1	1	1	1

4. Conclusions and Recommendations

This study has provided a comprehensive evaluation of regenerative suspension systems in electric vehicles, moving beyond simple theoretical estimates to a physics-based electromechanical coupling model. By integrating generator dynamics and regenerative damping feedback, we have established a more realistic framework to analyze the delicate balance between energy harvesting and ride comfort.

Our findings highlight that the energy recovery potential is deeply intertwined with road characteristics and vehicle speed. In scenarios involving discrete obstacles, the system achieved peak efficiency at low speeds, capturing 0.406 Wh for semicircular bumps at 2 km/h and 0.659 Wh for trapezoidal bumps at 5 km/h. While these results show that isolated impacts deliver high instantaneous power, their overall contribution to the energy budget is limited by the excitation's short duration.

The system's true potential, however, was revealed under continuous road excitations. Over a 1 km distance on a rough (ISO Class B) road, the system recovered a significant 70.4 Wh of energy at low speeds, while even a smoother (ISO Class A) surface yielded a substantial 57.1 Wh. We observed a clear inverse relationship between speed and energy recovery: as the vehicle accelerated to 50 km/h, recovered energy dropped to 3.95 Wh and 5.30 Wh on Class A and B roads, respectively. This decline underscores a critical physical insight: higher speeds reduce the time spent interacting with road irregularities, thereby limiting the window for energy harvesting.

From a ride comfort perspective, the introduction of regenerative damping force proved to be a double-edged sword. At lower speeds, the system naturally favors both high energy recovery and superior comfort, as vertical accelerations are lower. As speed increases, the rise in acceleration levels necessitates a strategic trade-off to maintain ride quality. Nevertheless, the fact that suspension travel remained within realistic mechanical limits across all tests validates the robustness of our electromechanical model.

It is believed that future studies examining the following situations will be beneficial in enriching the data obtained in this study and further contributing to the literature:

- New models can be tested that diversify road conditions and simulate higher cruising speeds.
- Vehicle weight distribution and suspension geometry need to be examined.
- Regenerative, active, and semi-active suspensions and predictive control models suitable for these systems can be developed to minimize the negative effect of braking on driving comfort and to increase energy recovery.

(iv) The effect of the energy gained from the regenerative suspension on the driving range should be evaluated based on the vehicle's annual usage period.

(v) Full vehicle models with advanced control systems, such as neuro-fuzzy control, can be created, with the powertrain also controlled, to ensure optimal energy consumption and driving comfort.

(vi) By integrating with autonomous vehicles, the suspension system can optimize driving style and favor paths that increase energy recovery.

Conflict of Interest Statement

The authors declare that there is no conflict of interest in the study.

CRediT Author Statement

Veli Gökhan Demir: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Visualization

Efe Savran: Conceptualization, Methodology, Supervision, Validation, Writing – review & editing

Declaration of Generative AI in the Writing Process

Artificial intelligence tools were used in this study solely for English grammar correction and to enhance the linguistic clarity of academic writing. AI was not used for content generation, data analysis, interpretation of the results, or formulation of scientific findings. All academic content, analyses, and conclusions presented in this manuscript are the author's work. The author takes full responsibility for the originality, scientific validity, and ethical integrity of this study.

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